Paper-1

Physics

13. An engine driver of a passenger train travelling at 40 m/s sees a goods train, ahead of his. The goods train is travelling in the same direction as the passenger train, with a constant speed of 20 m/s. The passenger train driver has reaction time of 0.5 sec. He applies the brakes which causes the train to decelerate at the rate of 1 m/sec^2, while the goods train continues with its constant speed. By what least distance (in m) last compartment of the goods train should be ahead of the engine of the passenger train in order to avoid the crash?

Ans. 00210.00

Sol. Before the applicator of brakes, the relative velocity of the passenger train with respect to goods trans $=40\text{-}20=20\,\text{m/s}$. The relative distance travelled by passenger train before travelles are applied during reaction time $20\times0.5=10\text{m}$

The relative acceleration of passenger train = (–1 – 0) = – 1 m/sec^2. The crash can only be avoided provided that after applying the brakes the relative velocity of the passenger train with respect to goods train becomes zero before the relative distance becomes zero.

Now applying the formula $v^2 = u^2 + 2\,a\,s$, we have $0 = (20)^2 - 2 \times 1 \times s$

or s = $\frac{(20)^2}{2}$ = 200 metre

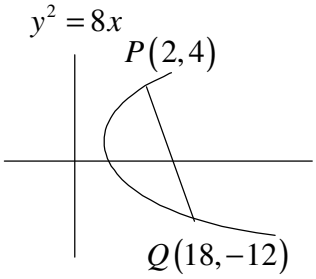
Total relative distance $200+10=210\text{m}$

Mathematics

49. The normal to the parabola $y^2 = 8x$ at the point (2, 4) meets it again at $(18,-12)$. If length of normal chord λ, then the value of λ^2 must be:

Ans. 00512.00

Sol. $y.y_1 = \frac{-y_1}{2a}(x - x_1)$



$\therefore$ equation of normal

$y - 4 = \frac{-4}{4}(x - 2)$

$y - 4 = -x + 2$

$x + y = 6$

$y^2 = 8x$

$y^2 = 8(6 - y)$

$y^2 + 8y - 48 = 0$

$y^2 + 12y - 4y - 48 = 0$

$y = -12, 4$

$x = 18,\ 2$

$\therefore PQ = \sqrt{(18-2)^2 + (-12-4)^2}$

$= \sqrt{16^2 + 16^2}$

$\lambda = 16\sqrt{2}$

$\lambda^2 = 512$

Paper-2

Chemistry

20. In order to increase the kinetic energy of ejected photoelectrons, there should be an increase in:

(A) Intensity of radiation

(B) wavelength of radiation

(C) Frequency of radiation

(D) both wavelength and intensity of radiation

Ans. C

Sol. KE_{max} of photo electron would increase on increasing the frequency of radiation or decrease in wavelength of radiation.

Mathematics

50. If α and β are the roots of the equation $x^2 + px + q = 0$ and also $x^{3900} + p^{1950}x^{1950} + q^{1950} = 0$ and if $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ are the roots of $x^n + 1 + (x+1)^n = 0,$ then the value of n must be equal to:

Ans. 01950.00

Sol. $x^2 + px + q = 0$

$\alpha + \beta = -p$

$\alpha\beta = q$

$x^{3900} + p^{1950}x^{1950} + q^{1950} = 0$

$\alpha^{1950} + \beta^{1950} = -p^{1950}$

$\alpha^{1950} \quad \beta^{1950} = q^{1950} \quad .....(2)$

$\frac{\alpha}{\beta} \rightarrow x^n + 1 + (x+1)^n = 0$

$\left(\frac{\alpha}{\beta}\right)^n + 1 + \left(\frac{\alpha}{\beta} + 1\right)^n = 0$

$\alpha^n + \beta^n + (\alpha + \beta)^n = 0$

$\alpha^n + \beta^n + (-p)^n = 0$

$\alpha^n + \beta^n = -(-p)^n \quad ....(3)$

$n = 1950$